NMC 2012 O-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn.1. Given that $p+q=0$ and $p \neq 0$, determine the value of $\frac{p^{2012}}{q^{2012}}+2$.

Qn.2. Given that $3^2+4^2+5^2+12^2=x^2+y^2$ is satisfied by only one pair of positive integers (x,y) with $x<y$, determine the value of $\sqrt{x+y}$.

Qn.3. A beginning algebra student mistakenly thinks that the equation $(x-2y)^2=x^2-4y^2$ holds for all values of x and y. Determine the value of $\frac{x}{y}$.

Qn.4. The number of fish that swam with me is the sum of digits of the largest integer x which satisfies the inequality $\frac{x}{x+1}<\frac{2008}{2009}$. How many fish swam with me?

Qn.5. Eugene, Godfrey and Fred enjoy different water sports. If Eugene sails, then Godfrey rows. If Eugene swims, Then Fred rows. If Godfrey does not swim, then Fred sails. Who does what?

Qn.6. A square of perimeter y is inscribed in a circle of circumference x. Find the ratio $x:y$.

Qn.7. Without using a calculator, find the value of

$$\frac{1234567890}{(1234567891)^2-(1234567890\times 1234567892)}$$

Qn.8. Let $P(N)$ and $S(N)$ denote the product and sum respectively of the digits of a positive integer N. For example, $P(63)=6\times 3=18$ and $S(63)=6+3=9$. Find all two digit numbers N for which $N=P(N)+S(N)$.

Qn.9. Triangle PQR has $\overline{PQ}=13, \overline{PR}=15$ and the altitude to side QR has length 12. Find all possible lengths of the side QR.

Qn.10. Determine the maximum possible value of the product of two numbers whose sum is 200.

SECTION B

Qn.11. (a) A jeweler has a 20 grams ring that is 60% gold and 40% silver. He melts it and adds enough gold to make it 80% gold. How many grams of gold should be added?

(b) If $S=1^2-2^2+3^2-4^2+\ldots+199^2-200^2$, determine the absolute value of S.

Qn.12. (a) Given a function F such that $F(0)=2$ and $F(x^2+1)=[F(x)]^2+1$ for all x, find $F(5)$.

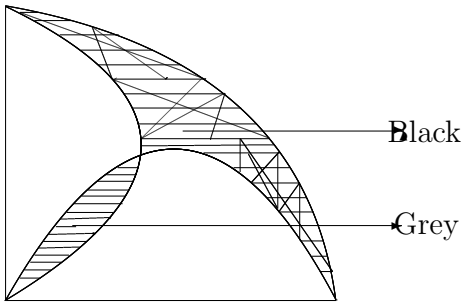
(b) In Ruti village, one-third of the children can play football, two-thirds can ride a bicycle and one-seventh can play football and ride a bicycle (though not necessarily at the same time). Given that there are fewer than 40 children in Ruti, how many of them can neither play football nor ride a bicycle?

Qn.13. (a) Evaluate $\left(1+\frac{1}{2}\right)\times\left(1+\frac{1}{3}\right)\times\ldots\left(1+\frac{1}{2006}\right)$.

(b) A bottle and a glass together hold the same amount of water as a jug. A bottle holds the same amount of water as a glass and a tankard together. Three tankards hold the same amount as two jugs. How many glasses would one tankard hold?

Qn.14. In a town, 10% of the dogs think that they are cats and 10% of the cats think that they are dogs. All the other cats and dogs are perfectly normal. When all the cats and dogs in a town are subjected to a test, 20% of the animals thought that they were cats. What percentage of them really were cats?

Qn.15. The digram shows a quarter circle with center O and two semicircles with diameters OA and OB. Calculate ratio of the area of the region shaded gray to the area of the region shaded black.



NMC 2012 O-LEVEL PAPER 1 SOLUTIONS

SECTION A

1. Given that $p+q=0$ implies that $p=-q$ and $\frac{p}{q}=-1$.

   Therefore,

$$\left(\frac{p}{q}\right)+2=(-1)^{2012}+2=1+2=3.$$

2. Using Pythagoras Theorem,

$$3^2+4^2+5^2+12^2=5^2+13^2=x^2+y^2.$$

   This implies that $x=5$ and $y=13$. Therefore, $\sqrt{x+y}=\sqrt{5+13}=\sqrt{18}=3\sqrt{2}$.

3.

$$\begin{aligned}(x-2y)^2 &= (x-2y)(x-2y)\\ &= x^2-4xy+4y^2\\ &= x^2-4y^2+4y^2-4xy+4y^2\\ &= x^2-4y^2-4xy+8y^2. \qquad (4)\end{aligned}$$

   The mistaken expansion has only the first two terms in equation (4). This therefore implies that $-4xy+8y^2=0$ or $4xy=8y^2$ or $y(4x-8y)=$ which implies that $y=0$ or $2y=x$.

   Therefore, $\frac{x}{y}=2$ if $y\neq 0$. Otherwise, it is not defined.

4.

$$\begin{aligned}\frac{x}{x+1}<\frac{2008}{2009} \quad &\Rightarrow \quad 2009x<2008(x+1)\\ &\Rightarrow \quad 2009x<2008x+2008\\ &\Rightarrow \quad x<2008.\end{aligned}$$

   Therefore $x=200$ and the number of fish which swam with me are $2+0+0+7=9$.

5. - If Eugene sails, then Godfrey rows. This means that Godfrey does not swim. If Godfrey does not swim, then Fred sails which is a problem, because if Eugene sails, then Fred cannot sail.
   - If Eugene swims then Fred rows. Since Godfrey does not swim because Eugene does, the Fred must also sail, which is not yet a solution.

- Hence Eugene rows. If Godfrey sails, then he does not swim; so Fred must also sail which is also a problem.

  Therefore, the only solution is that Eugene rows, Godfrey swims and Fred sails.

6. Let the side of the square be s and the radius of the circle be r as shown in Figure. Using Pythagoras Theorem, we have that

$$s^2 + s^2 = (2r)^2 = 4r^2; \;\Rightarrow\; s = \sqrt{2r}.$$

   Perimeter of the square is $y = 4s = 4\sqrt{2}r$ and circumference of the circle is $x = 2\pi r$. Therefore, $\frac{x}{y} = \frac{2\pi r}{4\sqrt{2}r} = \frac{\pi}{2\sqrt{2}r} = \frac{\pi\sqrt{2}}{4r}$. The required ratio is $x : y = \pi\sqrt{2} : 4r$.

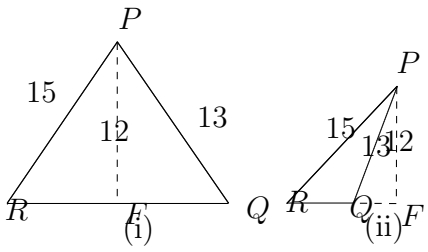
7. Let $n = 1234567890$. Then, $1234567891 = n + 1$ and $1234567892 = n + 2$. Therefore,

$$\begin{aligned}
\frac{1234567890}{(1234567891)^2 - (1234567890 \times 1234567892} &= \frac{n}{(n+1)^2 - n(n+2)} \\
&= \frac{n}{(n^2 + 2n + 1) - (n^2 + 2n)} \\
&= \frac{n}{1} \\
&= n = 1234567890.
\end{aligned}$$

8. Let the two digit number $N = 10a + b$ where a and b are digits. Then, $P(N) = ab$ and $S(N) = a+b$. Using $N = P(N)+S(N)$ we have that $10a+b = ab+a+b$ or $10a+b = (b+1)a+b$. Comparing coefficients we get $10 = b + 1$ implying that $b = 9$.

   Therefore, the numbers must have the second digit as 9. These are $19, 29, 39, 49, 59, 69, 79, 89$ and 99.

9. Let the foot of the altitude be F as shown in the Figure 15b.



   Using Pythagoras Theorem in $\triangle PRF$

$$\begin{aligned}
FR^2 &= PR^2 - PF^2 \\
&= 15^2 - 12^2 = 81.
\end{aligned}$$

Therefore, $\overline{FR} = 9$.

Likewise, using Pythagoras Theorem in $\triangle PQF$,

$$\begin{aligned} FQ^2 &= PQ^2 - PF^2 \\ &= 13^2 - 12^2 = 25. \end{aligned}$$

Therefore $\overline{FQ} = 5$.

The two figures depict the possible positions of F. Using (i), $\overline{QR} = \overline{QF} + \overline{FR} = 5 + 9 = 14$. The second possibility is depicted in Figure 15b(ii). Here $\overline{QR} = \overline{FR} - \overline{QF} = 9 - 5 = 4$.

10. Let the numbers be x and and y and let $p = xy$. We are given that $x + y = 200$ or $y = 200 - x$.

Method I.

$$\begin{aligned} p &= xy \\ &= x(200 - x) \\ &= -x^2 + 200x \\ &= -(x^2 - 200x) \\ &= -\left((x - 100)^2 + 100^2\right) \\ &= 10,000 - (x - 100)^2. \end{aligned}$$

The maximum value of p is $10,000$ and occurs when $x = 100$.

Method 2

Then $p = x(200 - x) = 200x - x^2 \Longrightarrow \frac{dp}{dx} = 200 - x$ and $\frac{d^2p}{dx^2} = -2$.

$\frac{dp}{dx} = 0$ when $200 - 2x = 0 \Longrightarrow x = 100$

When $x = 100$, $\frac{d^2p}{dx^2} = -2 < 0$ so a local maximum occurs at $x = 100$ and so the two numbers are $x = 100$ and $200 - 100 = 100$.The maximum possible value of the product is $100 \times 100 = 10000$

Section B

11. (a) Let x be the number of grams of gold to be added.
Gold concentration $= \frac{weight\ \ of\ \ gold}{total\ \ weight}$

$0.80 = \frac{20\times 0.6+x}{20+x}$
$12 + x = 16 + 0.8x \Longrightarrow 0.2x = 4$
Hence $x = 20$g

(b)

$$\begin{aligned} S &= 1^2 - 2^2 + 3^2 - 4^2 + \ldots + 199^2 - 200^2 \\ &= (1-2)(1+2) + (3-4)(3+4) + \ldots (199-200)(199+200) \\ &= -3 - 7 - 11 - \ldots - 399 \\ &= -1[3 + 7 + 11 + \ldots + 391 + 395 + 399]. \end{aligned}$$

Let

$$\begin{aligned} T &= 3 + 7 + 11 + \ldots + 391 + 395 + 399 \quad (5) \\ &= 399 + 395 + 391 + \ldots + 11 + 7 + 3 \quad (6) \end{aligned}$$

Summing the two equations gives

$$2T = 402 + 402 + 402 + \ldots + 402 = 100 \times 402 = 40200 \ \Rightarrow\ T = 20100.$$

Therefore,

$$S = -T = -20100 \text{ and } |S| = 20100.$$

12. (a)

$$\begin{aligned} F(1) &= F(0^2+1) = [F(0)]^2 + 1 = 2^2 + 1 = 5. \\ F(2) &= F(1^2+1) = [F(1)]^2 + 1 = 5^2 + 1 = 26. \\ F(5) &= F(2^2+1) = [F(2)]^2 + 1 = 26^2 + 1 = 677 \end{aligned}$$

(b) Let x be the number of children in the village. we are given that $\frac{1}{3}x$ can play football $\frac{2}{3}x$ can ride a bicycle and $\frac{1}{7}x$ can ride a bicycle and play football. This information implies that the number of children in the village must be a common multiple of 3 and 7. The only common multiple of 3 and 7 which is less than 40 is 21.

13. (a)

$$\begin{aligned} \left(1+\frac{1}{2}\right) \times \left(1+\frac{1}{3}\right) \times \ldots \left(1+\frac{1}{2006}\right) &= \frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \ldots \times \frac{2006}{2005} \times \frac{2007}{2006} \\ &= \frac{2007}{2} = 1003.5. \end{aligned}$$

(b)
- Let b be the capacity of the bottle in ml.
- Let g be the capacity of the glass in ml.
- Let j be the capacity of the jug in ml.
- Let t be the capacity of the tankard in ml.

We are given that

$$\begin{aligned} b+g &= j \\ b &= g+t \\ 3t &= 2j \end{aligned}$$

Therefore,

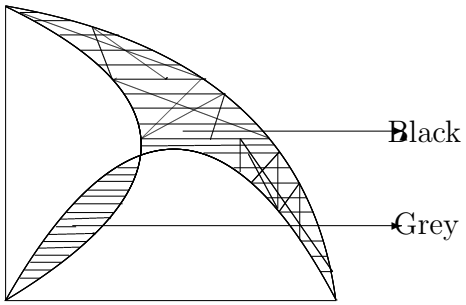
$$\begin{aligned} 3t &= 2j \\ &= 2(b+g) \\ &= 2(g+t)+2g \\ &= 2g+2t+2g = 4g+2t. \\ \Rightarrow\ t &= 4g. \end{aligned}$$

Therefore, four glasses would hold one tankard.

14. Let x represent the number of dogs and y represent the number of cats. We are given that $10\%x$ think that are cats and $90\%y$ think that they are cats. We are also given that $20\%(x+y)$ think that they are cats. So we solve,

$$\begin{aligned} 10\%x+90\%y &= 20\%(x+y) \\ \Rightarrow\ 10x+90y &= 20x+20y \\ \Rightarrow\ x &= 7y. \end{aligned}$$

Altogether the number of animals sum to $x+y=8y$ and the percentage of cats is $\frac{y}{8y}\cdot 100\% =$ 12.5%.



Let $OA = OB = r$. This implies that the area of quadrant $= \frac{\pi r^2}{4}$ and the area of one semi-circle $\frac{\pi r^2}{8}$. Therefore, $X+Z=\frac{\pi r^2}{8}$.

Area inside the quarter-circle but outside one semi circle is $= \frac{\pi r^2}{4} - \frac{\pi r^2}{8} = \frac{\pi r^2}{8}$ so $X+T=\frac{\pi r^2}{8}$. thus X +T=X +Z and this implies that T=Z so the areas of the shaded region are equal.

Thus the ratio of the area of the region shaded grey to the area of the region shaded black is $1 : 1$.